

```

#include <stdio.h>
#define MAX 5
int stack[MAX];
int top = -1;
void push(int value)
{
    if (top == MAX - 1)
        printf("Stack Overflow\n");
    else
    {
        stack[++top] = value;
        printf("Pushed %d to stack.\n",value);
    }
}
void pop()
{
    if (top == -1)
        printf("Stack Underflow\n");
    else
    {
        printf("Popped = %d\n", stack[top]);
        top--;
    }
}

```

```

int main()
{
    push(10);
    push(20);
    push(30);

    pop();
    pop();

    return 0;
}

```

```

#include <stdio.h>
#define MAX 50
int stack[MAX];
int top = -1;

void push(int value)
{

```

```

    if (top == MAX - 1)
        printf("Stack Overflow\n");
    else
    {
        stack[++top] = value;
        printf("%d pushed to stack.\n", value);
    }
}

```

```

void pop()
{
    if (top == -1)
        printf("Stack Underflow\n");
    else
    {
        printf("Popped = %d\n", stack[top]);
        top--;
    }
}

```

```

void display()
{
    if (top == -1)
        printf("Stack is empty.\n");
    else
    {
        printf("Stack elements:\n");
        for (int i = top; i >= 0; i--)
            printf("%d\n", stack[i]);
    }
}

```

```

int main()
{
    int choice, value;

    while (1)
    {
        printf("\n--- Stack Menu ---\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Display\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
    }
}

```

```

switch (choice)
{
    case 1:
        printf("Enter value to push: ");
        scanf("%d", &value);
        push(value);
        break;

    case 2:
        pop();
        break;

    case 3:
        display();
        break;

    default:
        printf("Invalid choice!\n");
}
}

return 0;
}

#include <stdio.h>
#define MAX 5

int queue[MAX];
int front = -1;
int rear = -1;

void enqueue(int value)
{
    if (rear == MAX - 1)
        printf("Queue Overflow\n");
    else
    {
        if (front == -1)
            front = 0;

        queue[++rear] = value;
        printf("Enqueued %d to queue.\n", value);
    }
}

```

```
    }  
}
```

```
void dequeue()  
{  
    if (front == -1 || front > rear)  
        printf("Queue Underflow\n");  
    else  
    {  
        printf("Dequeued = %d\n", queue[front]);  
        front++;  
    }  
}
```

```
int main()  
{  
    enqueue(10);  
    enqueue(20);  
    enqueue(30);  
  
    dequeue();  
    dequeue();  
  
    return 0;  
}
```